

Advanced (Carolina Reaper)

Open Ended Questions (FRQ)

Q9. Prove that $f(x) = 2x + 1$ has an inverse and find its inverse. Verify your answer using composition.

Q10. The cost C of producing x units of a product is given by $C(x) = 2x^2 + 5x + 100$. Find the inverse of $C(x)$ and interpret its meaning in the context of the problem.

Chef's Tips

- Tip 1: Make sure to check the work of your students carefully.
- Tip 2: Encourage your students to use the definition of an inverse function to verify their answers.
- Tip 3: Remind your students to label their graphs clearly and use the correct notation.